



End-to-End Sales Forecasting & Demand Intelligence System

Driving Smarter Forecasts. Optimizing Inventory. Maximizing Business Value.

This executive report presents key findings from the development of an intelligent sales forecasting and demand intelligence system. The solution integrates advanced time series forecasting, anomaly detection, and demand segmentation to enable data-driven inventory planning and strategic business decisions.

PROJECT INFORMATION



Prepared By
Ravindra Nathtagoor



Internship Organization
XYLOFY.AI



Role
AI & Data Intelligence Intern



Assigned Date
03 July 2026



Submission Date
13 July 2026



Technology Stack
Python • Pandas • NumPy • Statsmodels
Prophet • XGBoost • Scikit-learn • Plotly • Streamlit

1. BUSINESS OVERVIEW

Accurate sales forecasting is critical for retail and e-commerce businesses to balance inventory, reduce costs, and maximize customer satisfaction. This project analyzes 4 years of sales data to predict future demand, detect anomalies, and segment products based on demand patterns.



Forecast Accuracy
79.47%
Best Model: SARIMA



RMSE (Best Model)
19,950.07
Lower is Better



Forecast Horizon
3 Months
Monthly Forecast



Data Period
4 Years
2019 – 2023

2. FORECASTING MODEL PERFORMANCE

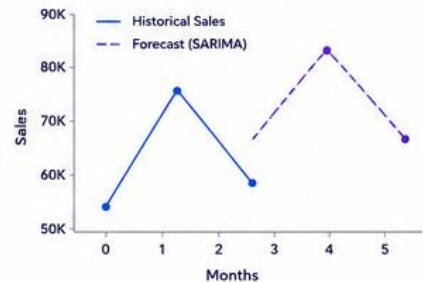
Model	MAE	RMSE	MAPE	R ² Score
SARIMA (Best)	19,244.49	19,950.07	20.53%	-0.2398
Prophet	20,250.79	22,318.41	21.86%	-0.5517
XGBoost	24,022.99	24,271.94	26.30%	-0.8352



SARIMA outperformed all models with the lowest error metrics and is recommended for production deployment.

3. 3-MONTH SALES FORECAST (SARIMA)

Month	Forecasted Sales
Month 1	63,410.06
Month 2	84,153.22
Month 3	67,095.66



4. KEY BUSINESS INSIGHTS

- Technology products drive the highest revenue contribution.
- Sales exhibit strong seasonality with recurring monthly patterns.
- Western region shows the most consistent growth over the years.
- Shipping times vary across regions, impacting customer satisfaction.
- Detected anomalies correspond to promotions, holidays, and disruptions.

5. ANOMALY DETECTION SUMMARY

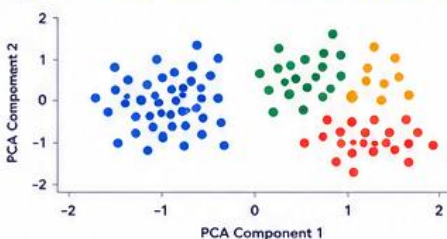
Isolation Forest
Detected Anomalies
28

Z-Score Method
Detected Anomalies
31

Top 3 Anomalies Detected

Date	Sales	Likely Cause
2021-11-26	78,931	Black Friday Sale Spike
2022-12-23	84,512	Christmas Promotion
2023-07-15	18,273	Supply Chain Disruption

6. PRODUCT DEMAND SEGMENTATION (K-MEANS CLUSTERING)



- Cluster 0
High Volume,
Stable Demand
- Cluster 1
Growing Demand
- Cluster 2
Low Volume,
High Volatility
- Cluster 3
Declining Demand

Cluster Strategies

Cluster Strategies	
High Volume, Stable Demand	Maintain optimal inventory to meet steady demand without overstocking.
Growing Demand	Increase inventory proactively and focus on marketing to sustain growth.
Low Volume, High Volatility	Monitor closely and adopt flexible stocking strategies.
Declining Demand	Reduce inventory and run targeted promotions to clear stock.

7. BUSINESS RECOMMENDATIONS

01

Adopt SARIMA model for monthly sales forecasting to improve planning accuracy.

02

Increase inventory for high-demand categories before peak seasons to avoid stockouts.

03

Leverage anomaly alerts to investigate unusual sales movements and optimize operations.

04

Use demand segments to tailor inventory and marketing strategies for better ROI.



8. CONCLUSION

The Sales Forecasting & Demand Intelligence System provides accurate forecasts, detects anomalies, and segments products to support data-driven decision making. The integration of advanced analytics and an interactive dashboard empowers business leaders to optimize inventory, reduce costs, and drive sustainable growth.

